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SonarCube Installation in Ubuntu

Step-1:- Install the PostgreSQL if it's not installed on your Ubuntu 24.04 workstation.

```
sudo apt install -y postgresql-common postgresql -y
```

Step-2:- Enable the PostgreSQL database server to automatically start at boot.

```
sudo systemctl enable postgresql
```

Step-3:- Start the PostgreSQL database server.

```
sudo systemctl start postgresql
```

Step-4:- Log in to the PostgreSQL database server as the postgres user.

```
sudo -u postgres psql
```

Step-5:- Create a new sonaruser PostgreSQL role with a strong password to use with SonarQube. Replace your_password with your desired password.

```
CREATE ROLE sonaruser WITH LOGIN ENCRYPTED PASSWORD 'your_password';
```

Step-6:- Create a new sonarqube database.

```
CREATE DATABASE sonarqube;
```

Step-7:- Grant the sonaruser role full privileges to the sonarqube database.

```
GRANT ALL PRIVILEGES ON DATABASE sonarqube TO sonaruser;
```

Step-8:- Switch to the sonarqube database.

```
\c sonarqube
```

```
GRANT ALL PRIVILEGES ON SCHEMA public TO sonaruser;
```

Step-9:- Exit the PostgreSQL database console.

```
\q
```

Install SonarQube

SonarQube is not available in the default package repositories on Ubuntu 24.04 and requires OpenJDK 21 to run. Follow the steps below to download the latest SonarQube release file and install SonarQube.

Step-1:- Update the server's APT package index.

```
sudo apt update
```

Step-2:- Install OpenJDK 17.

```
sudo apt install openjdk-17-jdk -y
```

Step-3:- Install Unzip to extract files from the SonarQube archive.

```
sudo apt install unzip
```

Step-4:- Verify the installed java version.

java -version

sudo wget

https://binaries.sonarsource.com/Distribution/sonarqube/sonarqube-25.2.0.102705 .zip

Step-5:- Extract files from the downloaded archive using Unzip.

unzip sonarqube-25.2.0.102705.zip

Step-6:- Move the extracted files to a systemwide directory such as /opt.

sudo mv sonarqube-25.2.0.102705/ /opt/sonarqube

Step-7:- Create a dedicated sonarqube system user without login privileges and a home directory.
sudo adduser --system --no-create-home --group --disabled-login sonarqube

Step-8:- Grant the sonarqube user full privileges to the /opt/sonarqube directory.

sudo chown -R sonarqube:sonarqube /opt/sonarqube

Install SonarScanner CLI

SonarQube uses code scanners depending on the target programming language to scan and analyze code quality. SonarScanner CLI is the default scanner if no specific scanner is specified on your system. Follow the steps below to install the SonarScanner CLI to analyze code on your workstation.
Step-1:- Visit the SonarScanner CLI page and verify the latest version to download. For example, run the following command to download the SonarScanner CLI version 7.0.1

Wget

https://binaries.sonarsource.com/Distribution/sonar-scanner-cli/sonar-scanner-cli7.0.1.4817-linux-x64.zip

Step-2:- Extract files from the archive depending on the downloaded version.

unzip sonar-scanner-cli-7.0.1.4817-linux-x64.zip

Step-3:- Move the extracted directory to /opt/sonarscanner.

sudo mv sonar-scanner-7.0.1.4817-linux-x64/ /opt/sonarscanner

Step-4:- Open the sonar-scanner.properties configuration file.

sudo nano /opt/sonarscanner/conf/sonar-scanner.properties

Step-5:- Find the following sonar.host.url directive and change the default

https://mycompany.com/sonarqube value to 127.0.0.1.

sonar.host.url=127.0.0.1

sudo chmod +x /opt/sonarscanner/bin/sonar-scanner

Step-6:- Link the sonar-scanner binary to the /usr/local/bin directory to enable it as a system-wide command.

sudo ln -s /opt/sonarscanner/bin/sonar-scanner /usr/local/bin/sonar-scanner

Step-7:- View the installed SonarScanner version.

sonar-scanner -v

Your output should be similar to the one below. 13:33:31.946 INFO SonarScanner CLI 7.0.1.4817 13:33:31.950 INFO Java 17.0.13 Eclipse Adoptium (64-bit) 13:33:31.951 INFO Linux 6.8.0-51-generic amd64

Configure SonarQube

SonarQube requires specific configurations for optimal performance, including database connections, Java runtime options, system resource limits, and user permissions. Follow the steps below to configure SonarQube to run on your server.

Step-1:- Open the main sonar.properties Sonarqube configuration file. sudo nano /opt/sonarqube/conf/sonar.properties

Step-2:- Add the following configurations at the end of the file. Replace sonaruser and your_password with actual PostgreSQL database user details.

sonar.jdbc.username=sonaruser sonar.jdbc.password=your_password sonar.jdbc.url=jdbc:postgresql://localhost:5432/sonarqube

sonar.web.javaAdditionalOpts=-server

sonar.web.host=0.0.0.0

sonar.web.port=9000

sudo nano /etc/sysctl.conf

Step-3:- Add the following directives at the end of the file.

vm.max_map_count=524288

fs.file-max=131072

sudo nano /etc/security/limits.d/99-sonarqube.conf

Step-4:- Add the following directives to increase the file descriptor and process limits for SonarQube. sonarqube - nofile 131072

sonarqube - nproc 8192

sudo ufw allow 9000/tcp

● Run the following command to install UFW and allow SSH connections if it's unavailable.

sudo apt install ufw -y && sudo ufw allow 22/tcp

Step-5:- Reload UFW to apply the firewall configurations.

sudo ufw reload

Step-6:- View the UFW status and verify that below are the only active firewall rules.

sudo ufw status

Your output should be similar to the one below: Status: active

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To Action From

-- ------ ----

22/tcp ALLOW Anywhere

9000/tcp ALLOW Anywhere

22/tcp (v6) ALLOW Anywhere (v6)

1. 9000/tcp (v6) ALLOW Anywhere (v6)

Set Up SonarQube as a System

Service Follow the steps below to set up a new system service for SonarQube to manage the application processes on your server.

Step-1:- Create a new sonarqube.service file.

sudo nano /etc/systemd/system/sonarqube.service

Step-2:- Add the following configurations to the file.

[Unit]

Description=SonarQube service

After=syslog.target network.target

[Service]

Type=forking ExecStart=/opt/sonarqube/bin/linux-x86-64/sonar.sh start
ExecStop=/opt/sonarqube/bin/linux-x86-64/sonar.sh stop

User=sonarqube

Group=sonarqube

PermissionsStartOnly=true

Restart=always StandardOutput=syslog

LimitNOFILE=131072

LimitNPROC=8192

TimeoutStartSec=5

SuccessExitStatus=143

[Install]

WantedBy=multi-user.target

sudo systemctl daemon-reload

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Step-3:-

Enable SonarQube to start at boot.

sudo systemctl enable sonarqube

Step-4:-

Start the SonarQube service.

sudo systemctl start sonarqube

Step-5:- View the SonarQube service status and verify that it's running.

sudo systemctl status sonarqube

sudo reboot now

GIT

version controller

this is the process of preserving older and later versions of the code.

All the developers upload their code into the version controlling system.

Git when working on the local machine uses 3 sections

1.working directory

2.staging area

3.local repository

1.**working directory**

Is the folder where the developer store or create his code.the files present here are initially call untracked files.

2.staging area

is the intermediate bufferZone where the files are initially moved into and these files are called stagged files

3.local repository

is the actual location where git performs version controlling.the files in the section are called committed files.

To move into the directory

$ cd C:

To create the directory

$ mkdir directoryname

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To create a file

$cat >filename

enter the data

press Ctrl+Z (or) $touch filename

To show the list of files

$ ls

$ ls -lrt

see the differences between normal files and text files

To send the files from working directory to staging area

$git add filename

To send multiple files from working directory to staging area

$git add filename1 filename2 filename3

To check the status

$ git status

it will show tracked and untracked files

To remove the files from stagging area

$ git reset filename

To send the files from staging area to localrepository

$ git commit -m "comments"

To see the status

$ git status it will show tracked and untracked files

To see the list of commits

$ git log

To see the list of commits in one line

$ git log –oneline